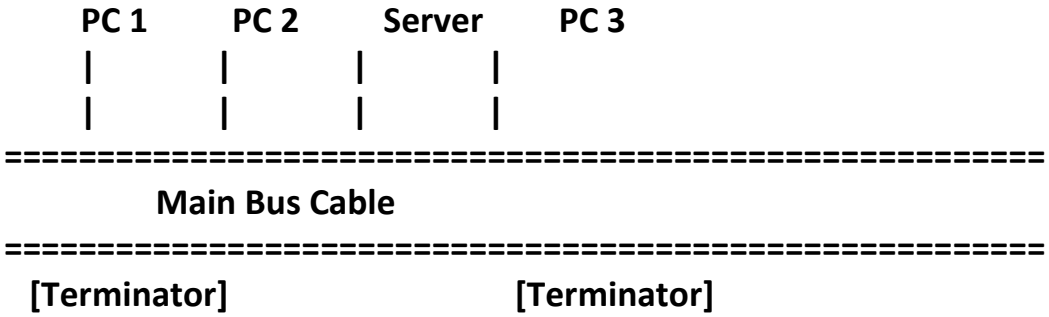
TASK: 1

Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.



Explanation:

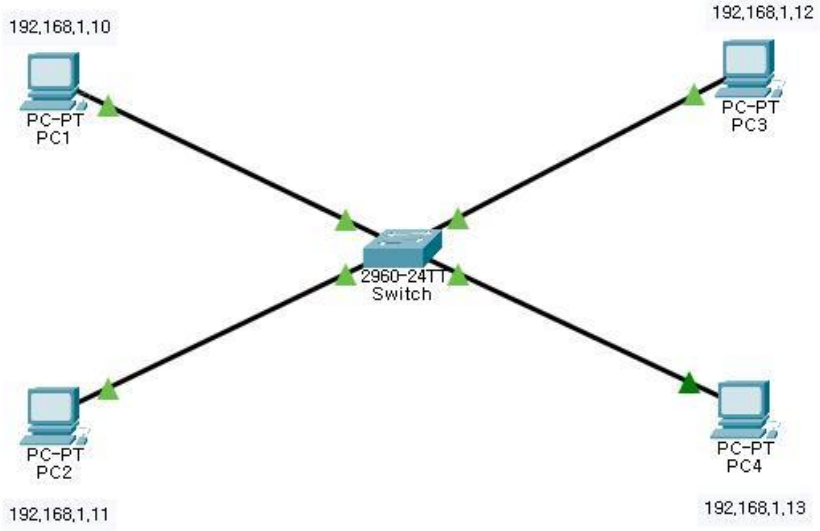
A bus topology is a network arrangement in which all devices are connected to a single main cable called the backbone. Terminators are placed at both ends of the cable to prevent signal reflections. If the main cable fails, the entire network may stop working.

Real-world example: Bus topology was used in older Ethernet networks in small offices and computer laboratories.

TASK: 2

In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

Star Topology Diagram

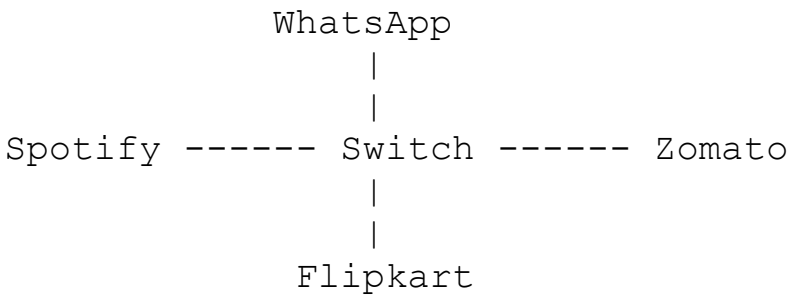


Steps to create the topology in Cisco Packet Tracer:

1. Open Cisco Packet Tracer.
2. Add one **2960 switch** to the workspace.
3. Add four PCs and arrange them around the switch.
4. Rename the PCs as follows:
    - PC0 – WhatsApp
    - PC1 – Zomato
    - PC2 – Spotify
    - PC3 – Flipkart
5. Select **Copper Straight-Through** from the Connections menu.
6. Connect each PC's FastEthernet0 port to a different Ethernet port on the switch.

7. Wait until the connection indicators turn green.
8. Take a screenshot of your completed topology and attach it to your assignment.

Expected topology:



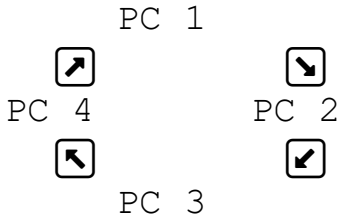
Result: Four PCs are connected to one central switch, forming a star topology.

TASK: 3

**Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Hint: Use arrows to show the direction of data flow.**

Ring Topology Diagram



Direction of data flow: Clockwise

Explanation:

In a ring topology, each device is connected to two neighboring devices, forming a closed loop. Data travels from one device to the next in the direction shown by the arrows until it reaches its destination. In a traditional ring network, a break in the ring can disrupt communication unless a protection or recovery mechanism is available.

Example: Ring topology has been used in some industrial networks and older Token Ring networks.

TASK: 4

Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Topology	Advantages (2)	Disadvantages (2)
Mesh Topology	1. Provides multiple paths for data, improving reliability. 2. A single link failure usually does not disconnect the entire network.	1. Requires more cables and network ports, increasing cost. 2. Installation and maintenance can be complex.
Star Topology	1. Easy to install, manage, and troubleshoot. 2. Failure of one connected cable usually affects only that device.	1. Failure of the central switch can disconnect all connected devices. 2. Requires more cable than a bus topology.